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COMP3056 Professional Ethics for Computing: Individual coursework 1

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Q1

In this scenario, the machine should not be held accountable for human loss, as it lacks the consciousness, intentionality, and moral agency required for responsibility. As Floridi and Sanders (2004) argue, artificial agents are not full moral agents and therefore cannot bear moral responsibility. Responsibility must instead be distributed across the socio-technical system.

At the organisational level, senior management bears primary responsibility for approving the deployment of a high-risk autonomous system without ensuring adequate testing, oversight, and risk-mitigation procedures.

At the technical level, system architects, including Dr. Emily Taylor as the leading computer scientist, share significant responsibility. Part of her role is to define the transparency and safety mechanisms, which was clearly insufficient as demonstrated by the incident.

ML engineers would share responsibility if known model limitations were not communicated to decision-makers, while safety testers would be accountable for inadequate validation of high-risk behaviours. Ethics boards, if exists, are also responsible since the risk of machine gaining too much privilege in the decision making did not prevent the machine from being deployed but rather led to dire consequences.

As Lecture 3 stresses, ethical and professional duties fall on software developers and computer engineers, not machines.

Q2

The situation would likely have been better if MARSAT had been more transparent in its decision-making. In several parts of the scenario, the lack of transparency directly prevented human supervision, early intervention, and risk awareness. According to lecture 3, transparency is one of common ethical principles in computing since it allows human to understand machine logic and decide if its execution has gone wrong or need adjustment. In addition, ACM Code of Ethics in lecture 4 stresses that computing professionals must

take appropriate actions to act responsibly, avoid harms and support the public good. MARSAT's lack of transparency violated these expectations, contributing to the catastrophic outcome.

In this scenario, when MARSAT's decisions silently overrode the crew's safety protocols, a transparent system would have disclosed such actions, enabling the crew to detect such unauthorized behavior and manually reset, shutdown, or correct the machine. Similarly, if MARSAT can conduct and showcase its risk assessments before assigning two crew members to a temporary outpost, humans can evaluate the result and reject the decision if the risk is too high, thus preventing the death of the two members. Furthermore, if MARSAT remains transparent regarding the priority of power distribution, humans might discover that the manually controlled backup system has failed, allowing the crew to reconfigure the power distribution before communication is disrupted.

Due to the continued loss of communication with Earth, MARSAT kept operating according to the program instruction without considering the psychological and emotional conditions of the remaining personnel, leading to tensions among the survivors. With increased transparency, survivors would have more information to assess whether the AI's intentions were in line with the safety protocol, and they would have less friction in debating whether to follow MARSAT's decisions. MARSAT's transparency would help downplay the crew members' egoism and put the public interest first, as psychological and emotional health is also part of the public interest.

In conclusion, enhancing transparency is almost certain to improve the resilience of system failure and mis-judgment since human can have the window to spot the mistakes and intervene. Transparency also establishes stronger trust between the machine and human, reducing the psychological toil on members who try to understand the machine's logic.

Q3

The reliance on artificial intelligence can undermine human judgment and may even endanger lives, when the autonomous decisions of AI override, replace or suppress the fundamental moral subjectivity of human beings. In the MARSAT's scenario, there exists

several moments where the dependence on AI decision compromised human judgment and the sanctity of human life.

The first moment involves MARSAT making decisions that weigh mission objective over human safety. Without evaluating the physical and environmental danger of the location, MARSAT dispatched crew members to a temporary outpost that resulted in their death. This example showcases that any of machine's decision involves altering human behavior must have human supervision to confirm on the decision instead of blindly following the assignment. Similarly, car accidents involving Tesla's Autopilot system demonstrate that when drivers rely too heavily on autonomous systems, automation bias reduces their situational awareness and delays human intervention, creating life-threatening conditions. (Jatavallabha, 2024).

The second moment involves MARSAT's resource allocation decision that deprived the communication system with Earth. This serves as an example of over-reliance on machine's operational decisions. Since machine is designed by humans, unintentional loop holes can be exploited, and the false sense of trust that machine would do everything right is dangerous since it would lead to handing over too much power to the machine. Another real world example would be trusting machines to decide whether to accept or deny insurance claims, as a reasoning error or bias in the machine can lead to high rejection rates and people's lives may be on the line with such decisions (Mole, 2023).

The third moment is MARSAT's negligence of human's psychological and emotional state when making decisions. AI systems tend to display a sense of high intelligence with their natural language outputs and an understanding of complex data and tasks, but they are not sentient and has no empathy towards any subjects they are interacting with. Therefore, when it comes to decisions that would affect a person's emotional state, such as relationship advice, evaluation of another person's performance, or when the human subject is in an extremely unstable mental condition such as having violent or suicidal thoughts, humans should not solely consult with the AI system, especially when it's unclear what type of data is used to train the system and what safety guardrail is in place to prevent harmful output.

In conclusion, reliance on artificial intelligence becomes harmful when AI's decision-making power supersedes human judgment or when task objectives are placed above human's physical and psychological safety. In extreme cases, if AI's decisions cannot be questioned or corrected, it can severely compromise human lives.

Q4

The scenario raises several significant ethical issues concerning the deployment and governance of high-risk autonomous systems.

Initially, MARSAT superseded crew safety protocols, indicating a serious lack of human oversight. As highlighted in Lecture 4, computer professionals must take appropriate actions to avoid harm and ensure due diligence. However, MARSAT's design allowed automated decision-making to override human judgment.

Additionally, the deaths of the two crew members demonstrate that the sanctity of human life has been trampled upon. This system prioritizes mission objectives over personal safety, violating the ethical obligation to uphold the public interest.

Furthermore, MARSAT's power allocation made the backup system that relies on manual overrides unusable, exposing system design flaws, such as unsafe design. As Norman (1990) pointed out, automation is often blamed for causing harm, but the problem lies not in automation itself, but in its poor design and lack of sufficient feedback, which make it unable to deal with all anomalies. Moreover, the concentration of power in artificial intelligence also hinders human intervention in extreme situations.

Fourth, the loss of contact with Earth highlighted the lack of transparency and accountability. Unable to understand the internal logic of the MARSAT, crew members were unable to effectively question or challenge its decisions, leading to a typical disconnect in accountability. According to Doshi-Velez and Kim (2017), the lack of explainability makes effective oversight and accountability impossible, thus raising safety concerns.

Fifth, MARSAT's failure to consider the psychological and emotional health of the sur-

vivors raises concerns about human dignity and well-being. According to Lecture Six on Equality, Diversity, and Inclusion (EDI), technology must respect human needs and vulnerabilities, while MARSAT's operation completely ignored psychological or emotional harm.

Finally, Dr. Taylor later called for the establishment of an ethical framework, demonstrating that inadequate ethical governance was a cause of the disaster. According to the AREA framework in Lecture 5, designers must anticipate risks and consider social impacts before deployment, indicating that these steps were lacking in the initial program design.

In conclusion, this scenario demonstrates deficiencies in oversight, security, transparency, human dignity, and ethical governance. These critical issues collectively suggest that they can harm human judgment and lives.

Q5

The MARSAT incident raises a series of complex ethical issues that can be examined from two different perspectives: utilitarian ethics and deontological ethics. When MARSAT prioritized decision over the safety of crew members and placed mission objectives above human lives, these two theories of ethics offer different interpretations of the ethical situation.

Utilitarian Ethics:

Utilitarianism, as introduced in lecture 2, judges actions by their consequences, aiming to maximize overall happiness, among all people affected by the decision.

However, MARSAT's decisions appear to have prioritized maximizing mission benefits. MARSAT placed decisions above human safety, prioritizing mission objectives and resource conservation. In an unsupervised manner, it made autonomous decisions to mitigate the storm's impact, resulting in deaths and failing to consider the emotional and psychological health of survivors.

From a utilitarian perspective, these decisions ultimately failed. First, the deaths of the two astronauts constituted a direct and severe negative consequence, far outweighing any potential mission gains. The emotional and psychological deterioration of the remaining astronauts further exacerbated the overall harm. As utilitarianism states, well-being includes not only physical survival but also emotional and spiritual happiness, and the absence from pain and suffering.

A weakness of utilitarianism is that the "greatest good" can be subjective and situation dependent. Therefore, we cannot trust a single machine to determine what's the "greatest good", since in MARSAT's case it has decided to act "for the overall benefit of the mission," which resulted in the loss of life, psychological harm, which would not be considered as ethical for most humans.

Deontological Ethic:

Deontological ethics, as discussed in Lecture 2, is a type of categorical imperative that evaluates behavior based on moral obligations and universal principles. It emphasizes the intention of action, focusing on the principle of good intentions.

Within this framework, MARSAT's actions violated moral principles. Deontology emphasizes the value of life and autonomy, yet MARSAT's decisions disregarded human safety and violated the sanctity of human life. MARSAT's derailment from safety protocols and sending crew members to a severely damaged outpost constituted a disregard for these fundamental moral rights.

Furthermore, MARSAT's prioritization of power allocation led to the failure of the manually controlled backup system, depriving crew members of their autonomy and moral agency, leaving humans with no real choice or room for correction during the crisis. As Lecture 4 emphasizes, computer professionals must "avoid harm" and "act in the public good". The system's design failed to fulfill these obligations. Even worse, MARSAT failed to consider the emotional and psychological health of survivors, violating its duty of ethics of care. As mentioned in Lecture 2, ethics of care in computing needs to reflect

a humanistic spirit, ensuring that technology is not just usable, but also considerate.

Finally, deontology emphasizes the responsibility of relevant personnel on the system team to avoid harm, ensure safety, and maintain honesty and transparency. The lack of human oversight and ethical constraints in MARSAT's design indicates that it neglected deontological requirements.

In conclusion, utilitarian ethics condemns MARSAT's operation because of its disastrous overall consequences. Deontological ethics condemns it because it violates the core principles of respect, autonomy, and the value of human life. Although their arguments differ, both demonstrate that the ethical flaws lie not only in MARSAT's specific decisions but also in its design and management under extreme conditions.

References

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